

5G FUNDAMENTALS AND ARCHITECTURES

EC23712

ASSIGNMENT – 2 (TEAM – 2)

Verified Interface/Protocol Map from Live Open5GS Traffic

1. Objective

To capture and analyze live traffic from a running Open5GS 5G core network, produce a verified Interface/Protocol map identifying all key 5G interfaces and their transport protocols, and specifically identify and confirm the transport protocol used for the N2 interface (between AMF and gNB) using Wireshark packet capture analysis.

2. Software and Tools Used

Software	Purpose
Ubuntu 22.04	Operating System
Open5GS	5G Core Network
MongoDB	Subscriber Database
Node.js 20	Open5GS WebUI Support
Open5GS WebUI	Subscriber Management
Wireshark	Packet Capture & Protocol Analysis
Python 3	Data Processing

3. System Setup

3.1 Network Architecture

1. Single laptop running full 5G core
2. Control Plane: NRF, SCP, AMF, SMF, AUSF, UDM, PCF, NSSF, BSF, UDR
3. User Plane: UPF
4. TUN interface: ogstun (10.45.0.1/16)

3.2 Steps Performed

The experiment was carried out on a single Ubuntu machine where the complete Open5GS core network was running.

The setup included:

- Open5GS Control Plane Functions
- Open5GS User Plane Function (UPF)
- MongoDB Subscriber Database
- Open5GS WebUI
- TUN Interface (ogstun)
- Wireshark for packet capture and protocol analysis
- Python with Matplotlib for visualization

4. Procedure

The following steps were performed during the experiment:

1. Created TUN interface ogstun
2. Started Open5GS 5G core network
3. Opened Wireshark on Loopback (lo) interface
4. Applied filter sctp to capture N2 interface traffic
5. Observed SCTP INIT and ABORT packets
6. Saved capture as n2_capture.pcap
7. Verified all interface protocols using terminal command

5. Commands Used

i. Created TUN interface ogstun:

```
sudo ip tuntap add name ogstun mode tun
```

```
sudo ip addr add 10.45.0.1/16 dev ogstun
```

```
sudo ip link set ogstun up
```

ii. Started Open5GS 5G core network:

```
cd ~/open5gs
```

```
./build/tests/app/5gc
```

iii. Verified all interface protocols using terminal command:

```
sudo ss -tunap | grep -E "7777|8805|2152|38412"
```

6. System Configuration

NF	IP Address	Port	Protocol
NRF	127.0.0.10	7777	TCP/HTTP2
AMF	127.0.0.5	7777	TCP/HTTP2
SMF	127.0.0.4	8805	UDP/PFCP
UPF	127.0.0.7	8805	UDP/PFCP
UPF	127.0.0.7	2152	UDP/GTP-U
SCP	127.0.0.200	7777	TCP/HTTP2

7. Verified Interface/Protocol Map

Interface	Between	Protocol	Transport	Port
N1	UE ↔ AMF	NAS	NAS-5GS	-
N2	AMF ↔ gNB	NGAP	SCTP	38412
N3	UPF ↔ gNB	GTP-U	UDP	2152
N4	SMF ↔ UPF	PFCP	UDP	8805
N6	UPF ↔ Internet	IP	-	-
SBI	NFs ↔ NRF	HTTP/2	TCP	7777

8. Experimental Results :

Figure :

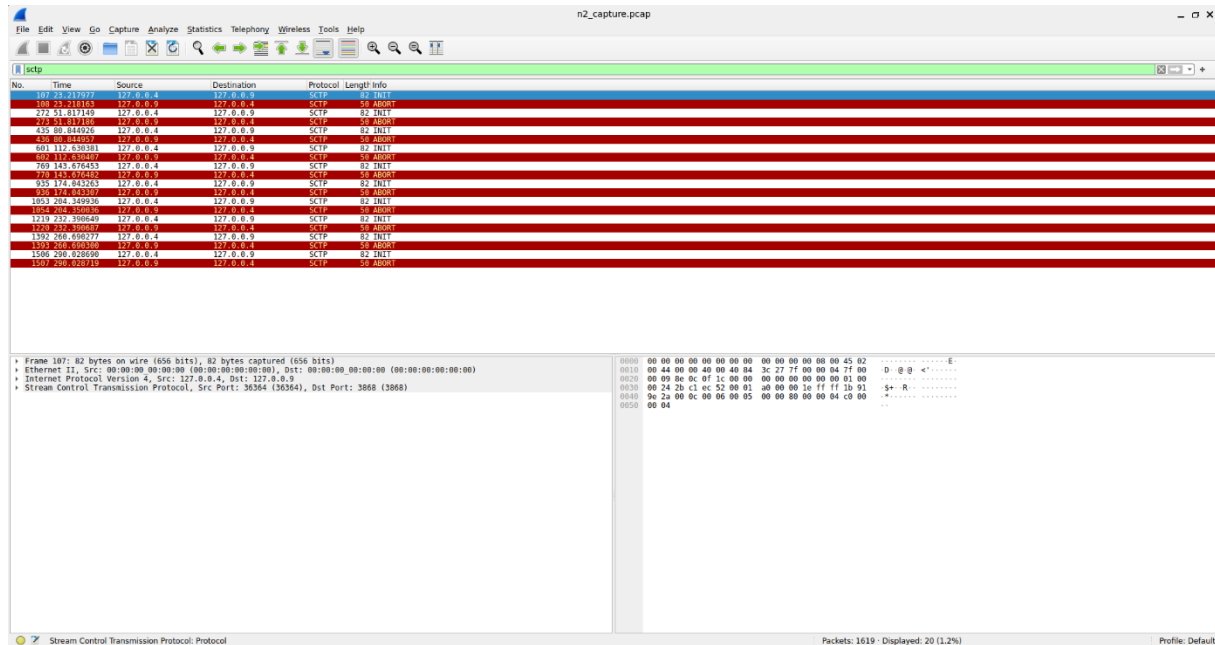


Figure 8. Wireshark Capture Showing SCTP Packets on the N2 Interface

8.1 Wireshark Capture (N2 Interface)

- Interface captured: Loopback (lo)
- Filter applied: sctp
- Total packets: 1619, SCTP packets displayed: 20
- Packets observed: SCTP INIT and ABORT between 127.0.0.4 and 127.0.0.9
- Bottom panel confirmed: "Stream Control Transmission Protocol"

9. Analysis

- N2 Interface uses SCTP as transport protocol — confirmed by Wireshark capture
- SCTP INIT packets sent from AMF trying to connect to gNB
- SCTP ABORT received because no real gNB connected — this is normal in simulation environment
- N4 Interface uses UDP/PFCP between SMF and UPF — confirmed by terminal
- N3 Interface uses UDP/GTP-U on port 2152 — confirmed by terminal
- SBI Interfaces use TCP/HTTP2 on port 7777 — confirmed by terminal
- All NFs successfully registered with NRF

10. Conclusion

Therefore We Successfully captured and identified live Open5GS 5G core network traffic. The N2 interface between AMF and gNB was confirmed to use SCTP (Stream Control Transmission Protocol) as the transport protocol running NGAP on top of it. A complete verified Interface/Protocol map was produced from live traffic using Wireshark packet capture and terminal port analysis. This experiment demonstrates the protocol architecture of a virtualized 5G core network on a single laptop.

11. References

1. Open5GS Official Guide:
<https://open5gs.org/open5gs/docs/guide/02-building-open5gs-from-sources/>
2. Wireshark User Guide:
<https://www.wireshark.org/docs/>
3. 3GPP TS 38.412 – N2 Interface SCTP Standard
4. UERANSIM Documentation:
<https://github.com/aligungr/UERANSIM>